

JAVA INHERITANCE PROGRAM

Application: Employee Salary | Program 1

1. Aim

To write a Java program using inheritance where the Manager class extends the Employee class and displays the employee salary, bonus, and total salary.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Program1.java**.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create an object of the Manager class.
6. The Manager class inherits the salary and showSalary() method from Employee.
7. Call showSalary() and showTotalSalary() to display the salary details.

3. Algorithm

1. Start the program.
2. Create the Employee parent class with salary and showSalary().
3. Create the Manager child class extending Employee.
4. Declare bonus in the Manager class.
5. Create showTotalSalary() to display bonus and total salary.
6. Create a Manager object in main().
7. Call showSalary() and showTotalSalary().
8. Stop the program.

4. Explanation of the Program

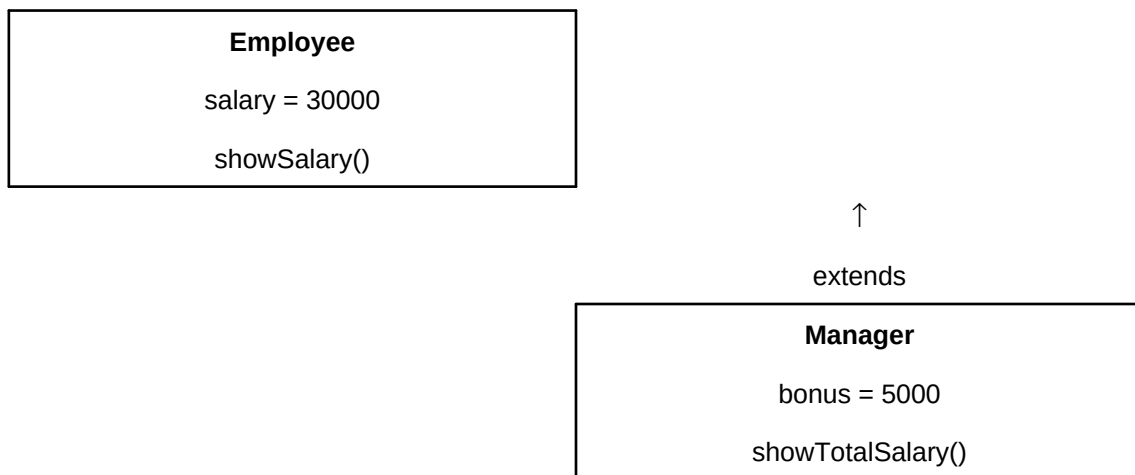
Employee class: It is the parent class containing the salary variable and the showSalary() method.

Manager class: It extends Employee, so it inherits salary and showSalary(). It also contains a bonus variable and showTotalSalary().

Inheritance: Manager inherits the properties and behavior of Employee using the extends keyword.

Manager object: The statement Manager m = new Manager(); creates an object of Manager, which can access both inherited and Manager-specific members.

5. Inheritance Diagram



6. Java Source Code

```
// Program 1: C1 extends C2
// Application: Employee Salary

class Employee {
    int salary = 30000;

    void showSalary() {
        System.out.println("Employee Salary = " + salary);
    }
}

class Manager extends Employee {
    int bonus = 5000;

    void showTotalSalary() {
        System.out.println("Bonus = " + bonus);
        System.out.println("Total Salary = " + (salary + bonus));
    }
}

public class Program1 {
    public static void main(String[] args) {

        Manager m = new Manager();

        m.showSalary();
    }
}
```

```
        m.showTotalSalary();  
    }  
}
```

7. Output of the Program

```
Employee Salary = 30000  
Bonus = 5000  
Total Salary = 35000
```

8. Result

Thus, the Java program successfully demonstrates inheritance by extending the Employee class with the Manager class and displaying the salary, bonus, and total salary.